

Radioactivity and Radioactive Decay Law



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Whatever we will discuss

- Radioactivity and its types
- Characteristics of radioactivity
- Radioactive decay law
- Important definitions

Radioactivity

The process of spontaneous emission of mysterious particles or rays from the temporary nucleus of a substance is called **radioactivity**.

The SI unit of radioactivity is Becquerel (Bq).

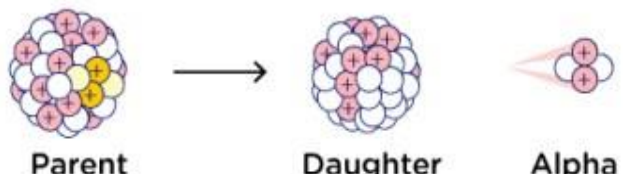
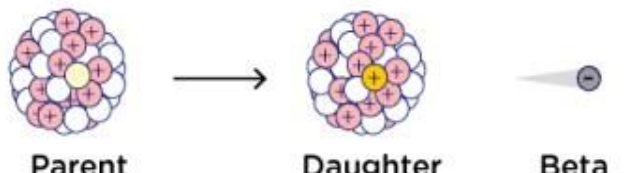
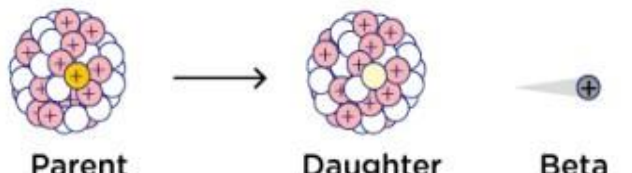
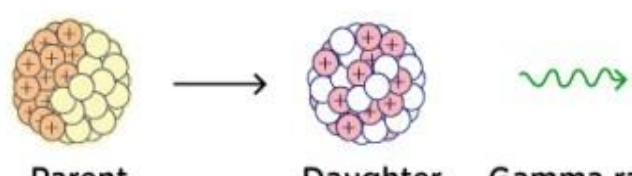
One radioactive decay per second is called one Becquerel.

Types of radioactivity-

- 1) Natural radioactivity
- 2) Artificial radioactivity

The substance from which these particles or rays are emitted is called the **radioactive substance**.

The rays which are emitted from the radioactive substance are known as the **radioactive rays**.

Decay Type	Radiation Emitted	Generic Equation	Model		
Alpha decay	${}^4_2\alpha$	${}_Z^AX \longrightarrow {}_{Z-2}^{A-4}X' + {}^4_2\alpha$	 Parent → Daughter Alpha Particle		
Beta- decay	${}^0_{-1}\beta$	${}_Z^AX \longrightarrow {}_{Z+1}^AX' + {}^0_{-1}\beta$	 Parent → Daughter Beta Particle		
Beta+ decay	${}^0_{+1}\beta$	${}_Z^AX \longrightarrow {}_{Z-1}^AX' + {}^0_{+1}\beta$	 Parent → Daughter Beta Particle		
Gamma emission	${}^0_0\gamma$	${}_Z^AX^* \xrightarrow{\text{Relaxation}} {}_Z^AX' + {}^0_0\gamma$	 Parent (excited nuclear state) → Daughter Gamma ray		

Characteristics of Radioactivity

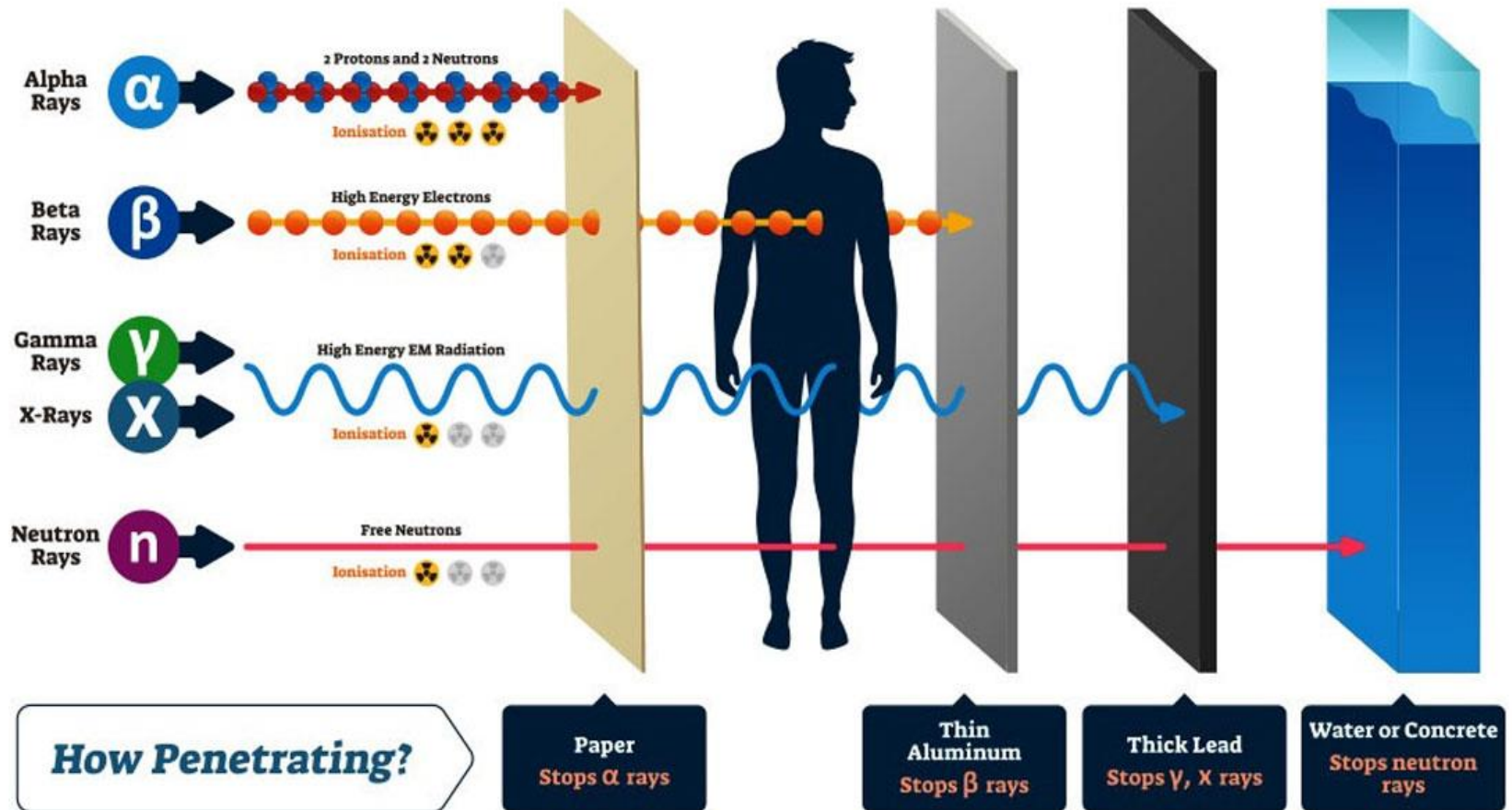
The following characteristics are observed for radioactivity:

- ✓ The elements whose atomic number is greater than 83, they exhibit radioactivity.
- ✓ Radioactivity is an irreversible and spontaneous nuclear phenomena.
- ✓ Change of temperature or pressure, surrounding radiations, electric or magnetic fields, external force etc. do not influence radioactivity.
- ✓ Normally alpha, beta and gamma rays are emitted from radioactive substance.

Properties of alpha, beta and gamma rays

Properties	α rays	β rays	γ rays
What are they?	Helium nucleus (${}_2\text{He}^4$) consisting of two protons and two neutrons.	They are electrons (${}_{-1}\text{e}^0$), basic elementary particle in all atoms.	They are electromagnetic waves consisting of photons.
Charge	Positively charged particles. Charge of each alpha particle = $+2e$	Negatively charged particles. Charge of each beta particle = $-e$	Neutral particles. Charge of each gamma particle = zero
Ionising power	100 time greater than β rays and 10,000 times greater than γ rays	Comparatively low	Very less ionization power
Penetrating power	Low penetrating power (even stopped by a thick paper)	Penetrating power is greater than that of α rays. They can penetrate through a thin metal foil.	They have a very high penetrating power greater than that of β rays. They can penetrate through thick metal blocks.
Effect of electric and magnetic field	Deflected by both the fields. (in accordance with Fleming's left hand rule)	Deflected by both the fields; but the direction of deflection is opposite to that for alpha rays. (in accordance with Fleming's left hand rule)	They are not deflected by both the fields.
Speed	Their speed ranges from 1/10 to 1/20 times the speed of light.	Their speed can go up to 9/10 times the speed of light.	They travel with the speed of light.

TYPES OF RADIATION



Radioactive decay law

Statement:

At any moment the number of radioactive atoms that disintegrate in unit time is directly proportional to the number of unchanged radioactive atoms remaining.

$$-\frac{dN}{dt} \propto N$$

Explanation:

If the rate of radioactive disintegration of atom is $\frac{dN}{dt}$ and if N is the number of

unchanged atoms at time t , then $-\frac{dN}{dt} \propto N$

$$\text{or, } -\frac{dN}{dt} = \text{constant} \times N$$

$$\text{or, } \frac{dN}{dt} = -\lambda \times N$$

$$\text{or, } \frac{dN}{N} = -\lambda \times dt \quad \dots \quad \dots \quad (1)$$

Here, λ is the radioactive decay constant. Since the number of atoms of radioactive substance decreases, hence a negative sign has been used in the equation.

Radioactive decay law

Integrating the equation (1) we get,

$$\log_e N = -\lambda t + C \quad \dots \quad \dots \quad (2)$$

Here C is the constant of integration. Its value is to be determined.

Suppose at the initial time, $t = 0$, then $N = N_0$. Thus, from equation (2) we get

$$\log_e N_0 = C \quad \dots \quad \dots \quad (3)$$

Now, from equations (2) and (3) we get

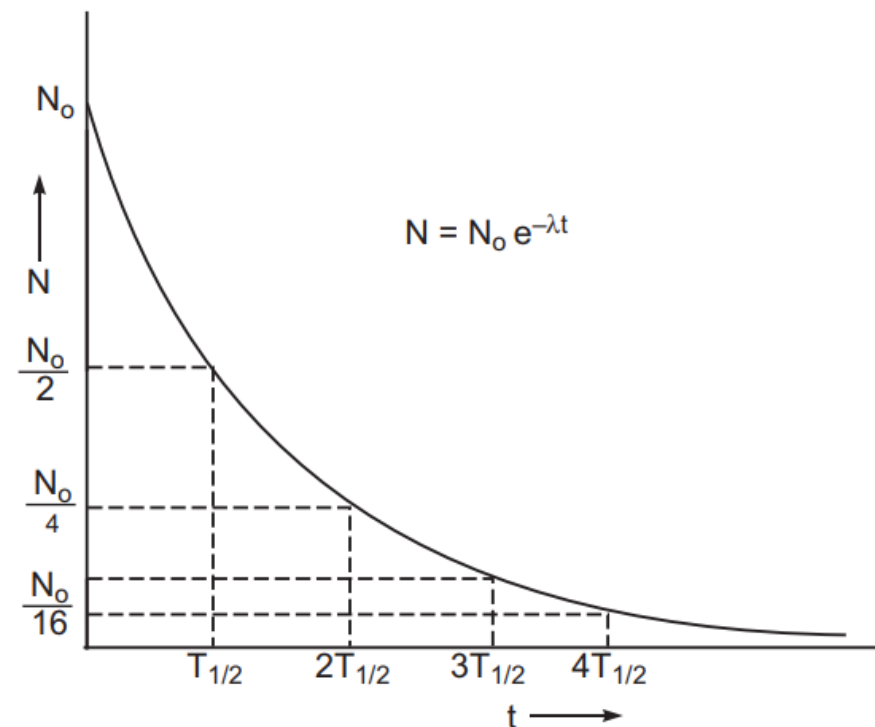
$$\log_e N = -\lambda t + \log_e N_0$$

$$\text{or, } \log_e N - \log_e N_0 = -\lambda t$$

$$\text{or, } \log_e \frac{N}{N_0} = \log_e e^{-\lambda t}$$

$$\text{or, } \frac{N}{N_0} = e^{-\lambda t}$$

$$\therefore N = N_0 e^{-\lambda t}$$



This is the law of radioactive decay or disintegration.

Important Definitions

Radioactive decay constant

The decay constant may be defined as the ratio of the number of atoms which disintegrates in unit time to the number of atoms present.

$$\lambda = \frac{-dN/dt}{N}$$

Half-life

Half-life is defined as the time in which the radioactive atoms are reduced to half their initial amount.

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$$

Mean life

The mean life of a radioactive element is defined as the ratio of the total life time of all the radioactive atoms to the total number of such atoms in it.

$$\tau = \frac{1}{\lambda} = \frac{T_{1/2}}{0.693}$$

Radioactivity related mathematical problems

[Physics for Engineers (Part-2) by Dr. Gias Uddin Ahmad (pp. 707-712)]

Ex. 16.1, Ex. 16.2, Ex. 16.3, Ex. 16.4, Ex. 16.5 & Ex. 16.6



Thank You
Thanks for your attention